

Scanner Specifications

Minimum Host Computer Specifications

- Pentium IV or AMD Athlon CPU 1.7GHZ or higher
- 1 40GB hard drive to be used for the operating system, application programs, and storage. If IDE, it must be configured as primary master.
- 1 40GB hard drive capable of a sustained transfer rate of 30MB/sec. This is to be a dedicated drive for the scan data buffer. If IDE, it must be a secondary master drive. Drive letter must be N:\ and **it must NOT be formatted**. This should be a 7200rpm drive. **See the “Disk Management” chapter for more information on configuring this drive.**
- Windows 2000, Service Pack 2
- 512MB of RAM
- A motherboard with an integrated USB 2.0 controller as part of the motherboard chipset. A dedicated port is recommended. USB 2.0 hubs have not been fully tested for reliability.
- Separate AGP Graphics card. Using a motherboard with a built-in graphics card will be problematic.
- Motherboards with the following chipsets are recommended:

Recommended Motherboards and Chipsets
Intel 845E
Intel 845GE
Intel 845G
Intel 845PE
Intel 845E
SIS 651
SIS 962
SIS 645DX
SIS 648
VIA VT8235 integrated USB controller

- Microsoft USB 2.0 drivers.
- Integrated video cannot be used. Use an add-on AGP video card.
- SCSI CD-RW drive. (Plextor 12/10/32S recommended.)
- 1.5GB Pagefile.(Virtual Memory)

Power

- Auto-switching power supply for 100-240VAC
- 50-60Hz Input voltage frequency
- Standard 3 conductor IEC 320 male power cable connector

Contents of Box

- F-235 Film Scanner
- 6ft. USB 2.0 Cable
- Software and User Manual CD
- Power Cord
- Region Specific accessory pack, if applicable.

Size

- 17.3"W x 13.6"H x 12.6"D with cover on

Light Source

The F-235 uses a 50 Watt Solux Halogen light bulb. During scans, the bulb operate at standard voltage. After completing a scan operation, the bulb power will be reduced to a lower voltage to increase the bulb's life expectancy.

Film Types and Variations

- The F-235 is designed to scan film in 35mm and 24mm (APS) formats. Color negative, color reversal (positive or slide) film, black and white, C-41 processed black and white films are all supported.
- APS film can only be scanned once removed from the canister, but cannot be cut into strips. All APS formats (C, H, and P) will scan only in full frame (H) format.
- 35mm cut strips are supported in sizes from 2 frames to 41 frames. However, to ensure DX code reading, the strips must be a minimum of 3 frames.
- Mounted slides are not supported.

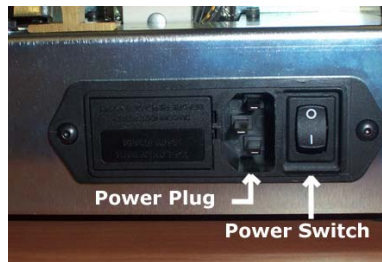
DX Code Reading

- 35mm DX codes are read using the ISO 1007 specifications and are used by the host manager.
- APS encoding will only be read by a full roll of film, and will not be read from an APS strip.

Installing the F-235

The F-235 is a peripheral device. It cannot operate apart from a computer that meets the required specifications listed on page 2 of this manual.

1. Clear a space near your computer system to place the F-235 scanner. The USB 2.0 cable must be able to reach the PC, so it cannot be more than 6 feet (2m) from the computer. There must also be 4 inches (10cm) clearance above and behind the scanner's ventilation duct.
2. Connect the power cord to the power supply plug on the back of the scanner.



3. Connect the USB 2.0 cable to the back of the scanner.



4. Connect the USB 2.0 cable to the back of the computer, in a USB 2.0 port. **Do not plug in to a USB 1.0 connector!**

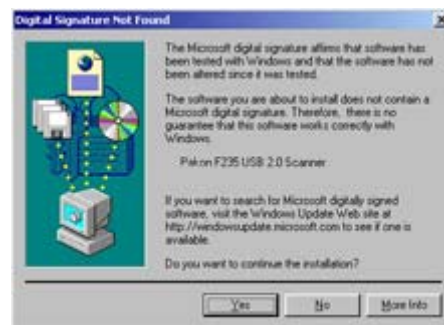


5. Insert the F235 software CD. Run 'setup.exe' from the CD-ROM drive.
6. While computer is on and Windows 2000 is loaded, turn the scanner power switch into the 'on' position.

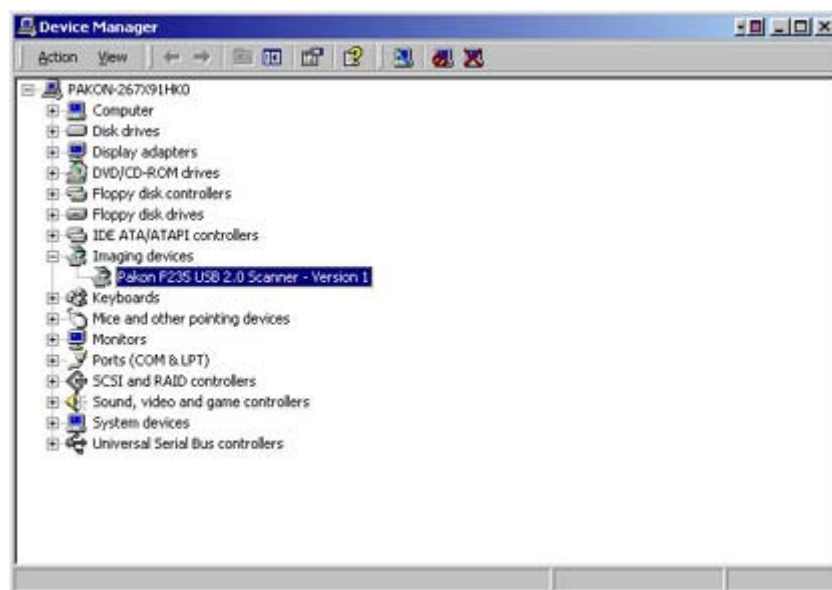
7. Insert F-235 Software CD. The computer will prompt the user to locate the driver software. This driver software is located on the CD in \Program Files\Pakon\F235Driver. This driver is also on the C:\ if the software has already been installed.



8. The install will ask for authorization to install the driver despite it not having been “digitally signed.” Yes must be selected for the driver to install.



9. After the driver is finished installing, verify that the scanner is recognized by viewing the scanner properties in the Device Manager under the heading, “Imaging Devices.”



BIOS Configuration

The following BIOS options are used on systems configured by Pakon, and are recommended to be configured as outlined below:

Option	Setting
Plug and Play	Disable
AGP Aperture Size	32
PCI Latency Timer	255
ACPI Suspend State	S1 State
USB Boot	Disable
PXE Boot to Lan	Enable
PCI to DRAM Prefetch	Disable

SCSI BIOS Configuration

These settings should be followed in the SCSI BIOS of the SCSI card used in the Pakon computer system.

Option	Setting
Sync. Transfer Rates	10
Bootable CD-ROM	Disable
BIOS Support for Int. 13 Ext.	Disable

Operational Information

Operation

When scanning film, ensure the appropriate film type is selected. Do not attempt to scan 35mm film when APS is selected, and likewise do not scan APS film when 35mm film has been selected.

Insert the film emulsion side-up, with the first frame first on the right side of the scanner. This will ensure that the DX codes are read and that the framing will be done correctly.

Do not attempt to remove film from the scanner while the scanner motor is moving! This will cause damage to both the film and the scanner.

Calibration

There are two types of calibration for the F-235.

Color calibration is set during the manufacturing process, as well as any service that involves the CCD or any optical filter. This is to be performed by trained and authorized service personnel only.

Scanner corrections are performed automatically. They consist of a series of steps performed when the scanner is first turned on, as shown below:

- Start-up corrections
- Initial bulb warm-up
- Gain and exposure Control Corrections
- Run Time Corrections

Format Change

The scanner will automatically change the film guide, lens position, CCD position, or color filter position whenever one of the following steps take place:

- Film Format change between 35mm and APS
- Film type change between color negative, color reversal, and black and white
- Magnification Change (Change of scanning resolution)

Film Gate Change

The film gate opens and closes depending on the size of the film being scanned. It will open for 35mm film and close for APS film. This gate will open or close automatically when a different film size is selected

Post Warm-Up

The steppers will take less than 20 seconds to move the lens and CCD into proper position for the selected focus position.

LED Indicator Lights

Power LED	Function
Solid Green	+5V is functioning
Off	+5V is not functioning

Status LED	Function
Solid Green	Scanner Ready
Blinking Green	Scanner is Scanning
Blinking Yellow	Scanner is Calibrating
Blinking Red	Scanner Error
Off	Scanner not Functioning

APS LED	Function
Solid Green	Scanner in APS mode
Blinking Green	Scanner is Scanning
Blinking Red	APS mode Error
Off	Scanner in 35MM mode

Digital ICE

Digital ICE is designed to assist in removing scratches, dust, and debris artifacts from scanned film. It should be used as a complement to a photo lab's normal cleaning procedures. Photo lab's are still advised to clean the floors and work surfaces regularly.

It is also recommended to clean all film with a lint-free cloth before attempting to make prints 8"x10" or higher.

If a lab is having persistent dust problems, it is recommended to perform the following regularly:

- Clean all customer film with a lint-free cloth before scanning it.
- When the scanner is turned off, cover it with a plastic, or lint-free cover.
- Use a lint-free cloth daily to clean the cover of the scanner and the surface of the table it is positioned on.
- Clean or replace the air filter on the back of the scanner once per week.

Operator Maintenance

Warning: These procedures require the operator to remove up to two of the three covers on the scanner. Do not attempt to operate the scanner without these covers properly in place!

WARNUNG: Bei diesen Verfahren muss der Bediener bis zu zwei der drei Abdeckungen am Scanner entfernen. Der Scanner darf nicht betrieben werden, wenn diese Abdeckungen nicht ordnungsgemäß angebracht sind!

ATTENTION : au cours de ces procédures, vous devez ôter deux des trois couvercles du scanner. N'essayez pas de faire fonctionner le scanner sans que ces couvercles de protection soient correctement à leur place !

Dust cleaning of illumination

The F-235 is a precision optical device. A relatively smoke and dust free environment is necessary. The scanner must be cleaned periodically to remove large collections of dust or debris inside the scanner, on the illumination. **This procedure is recommended to be performed every two weeks.** Software is designed to compensate for smaller collections of dust or debris, but cleaning may be required. Digital ICE is only designed to complement the lab's normal cleaning procedures--which should include cleaning the floor and work surfaces regularly.

Follow each step by a verification process to determine if the cleaning was effective. If the verification determines that level of cleaning was ineffective, the next level is required.

1. Simple cleaning with compressed air and or lens brushes.
2. Cleaning with rubbing alcohol and a cotton swab.
3. Service call.

Cleaning the Illumination with Compressed Air

1. Turn off the scanner.
2. Disconnect the USB and power cables.
3. Gently remove the side panel of scanner, by pulling back on it. It should snap out of place.



Do not touch anything inside the scanner while the power is on. The cover is only off to allow dust and debris to be blown out of the scanner.

WARNUNG: Berühren Sie keine Bauteile innerhalb des Scanners, während das Gerät eingeschaltet ist. Die Abdeckung wird nur deshalb abgenommen, damit Staub und Verschmutzungen aus dem Inneren des Scanners heraus geblasen werden können.

ATTENTION : ne touchez aucun élément interne du scanner lorsque ce dernier est sous tension. Le couvercle est ouvert pour permettre d'éliminer poussière et débris du scanner.

4. Connect power to the scanner.
5. Turn the scanner on.
6. Remove the small film path cover.



7. Aim the hose of a can of compressed air into the film path--aiming for the clear piece of glass in the middle.

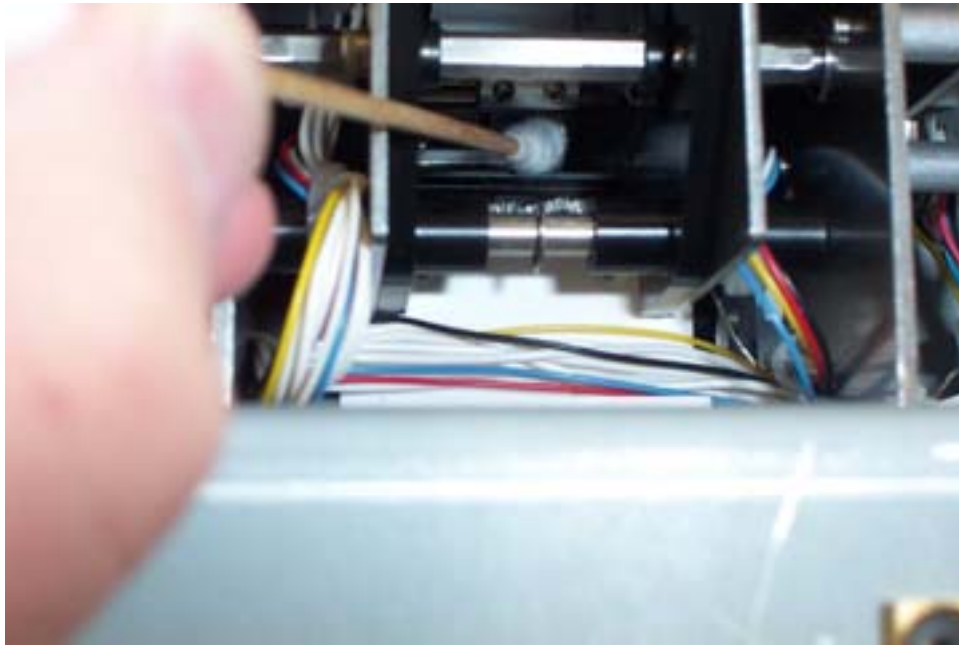


8. It may be beneficial to insert a strip of film into the scanner before blowing it out. This will focus all the air onto the light transfer bar, and should dislodge any dust or debris.
9. When finished, put the film path cover back in place.
Turn off the scanner immediately when finished cleaning the scanner!
10. Turn off the scanner and unplug the power cable.
11. Replace the side cover by snapping into place.
12. Plug in the USB and power cables.

Cleaning Illumination with Cotton Swab

CAUTION--Be Careful not to scratch or damage the top of the glass light bar!

1. Turn off the scanner.
1. Disconnect the USB and power cables.
2. Gently remove the side cover.
3. Using a long cotton swab, dabbed in isopropyl alcohol, clean the illumination bar inside the film track assembly.
4. The illumination bar is a small piece of glass, sitting upright.



5. After cleaning the illumination bar, replace side cover and turn on.

Light Bulb Replacement

The light bulb will have a nominal life of 100 working days or longer, depending on certain variables. This number is based on the following assumptions:

- 6 working days per week
- 10hours per day

Changing the light bulb does not require any special tools.

Replacing the Scanner Light Bulb

CAUTION: The lamp may be hot. Allow the bulb to cool before attempting to replace it. Use gloves when replacing the scanner light bulb. Fingerprints can damage the light bulb.

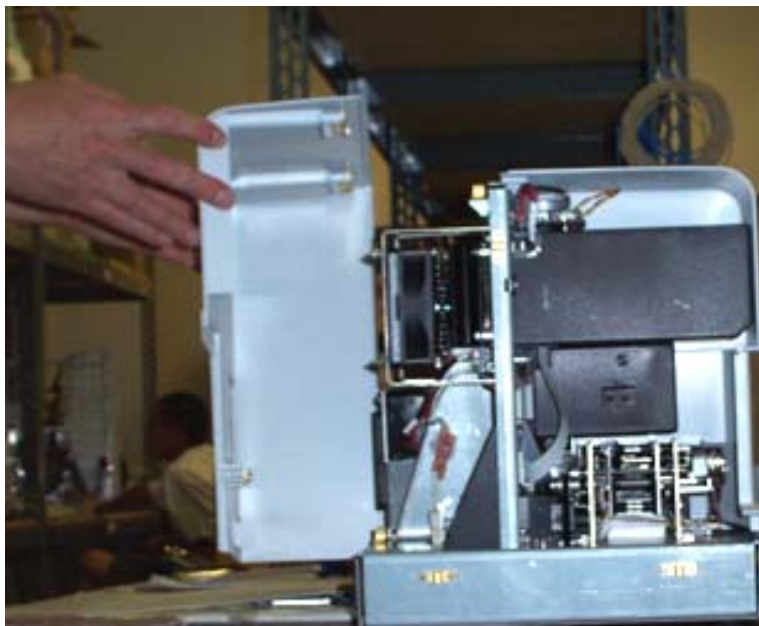
VORSICHT: Die Lampe kann heiß sein. Lassen Sie die Glühlampe abkühlen, bevor Sie sie austauschen. Berühren Sie die Scanner-Glühlampe nur mit Handschuhen. Fingerabdrücke können die Glühlampe beschädigen.

ATTENTION : l'ampoule risque d'être chaude. Attendez qu'elle refroidisse avant d'essayer de la remplacer. Utilisez des gants pour remplacer l'ampoule du scanner. Les empreintes digitales peuvent l'endommager.

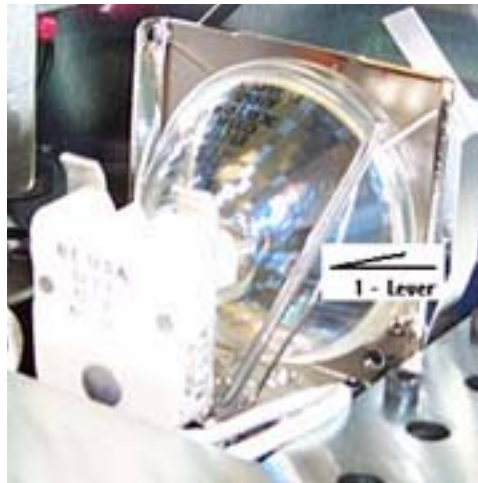
1. Close all scanning software.
2. Turn off the scanner.
3. Disconnect the USB and power cables.
4. Gently remove the side cover by pulling on it. It should snap out of place.



5. Gently pull off the back cover. It should snap off much like the side cover.

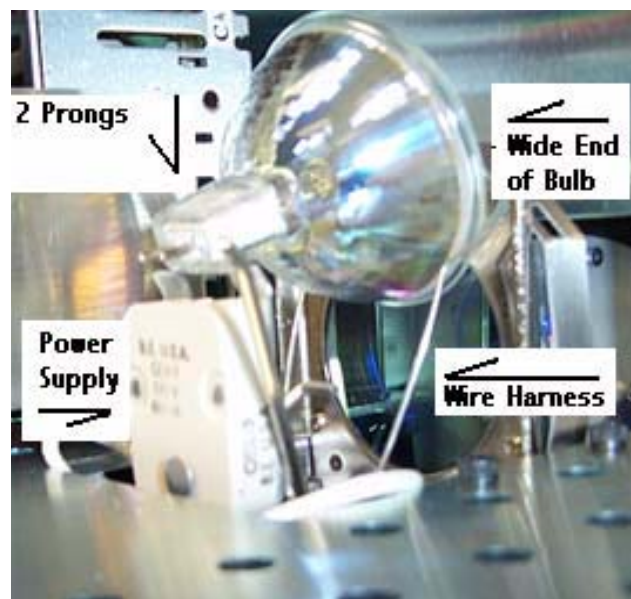


6. Locate the small lever, just to the back of the bulb.



Gloves must be worn to ensure the bulb is not damaged by dirt or skin oils.

7. With one hand, gently pull the lever towards the back of the scanner.
8. As the lever is pulled, the bulb will be pushed up, and partially out of its wire harness.
9. Once the bulb has been pushed up far enough, use your other hand to gently remove the bulb from the harness.
10. Orient the new light bulb so that the largest, or widest part of the bell shape is facing the front of the scanner. And place the smaller end with the prongs, facing the back.
11. Rest the wide end of the bulb on the top edge of the wire harness, do not push or force it downward.
12. Spin or rotate the bulb so that the two prongs line up with the grooves in the lamp ballast.



13. Gently apply downward pressure equally to both the front and the back of the bulb.
14. Slide the bulb into place, until it fits securely.
15. Replace the back cover by snapping it into place.

16. Replace the side cover by snapping it back into place.
17. Plug in USB and power cables.
18. Turn scanner on, and allow to warm up before using again.

Cleaning the Fan Filter

The fan filter collects dust from inside the scanner. This filter needs to be cleaned every two weeks.

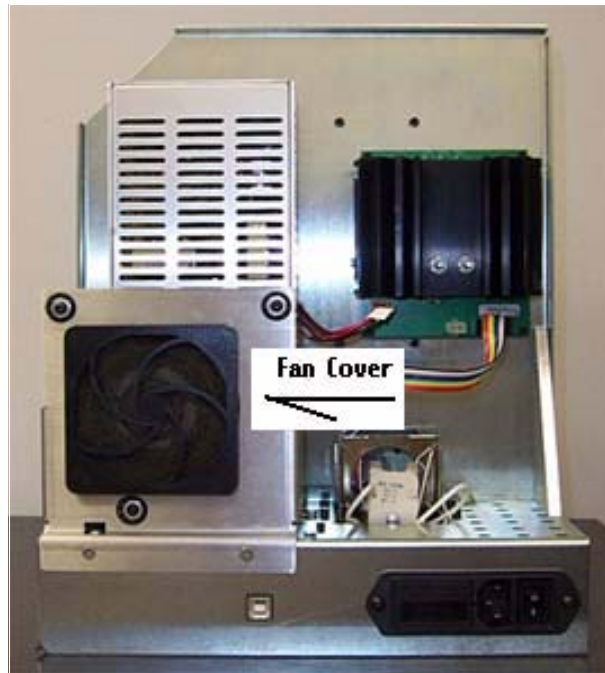
1. Close scanner software.
2. Turn off the scanner.
3. Disconnect USB and power cables.
4. Gently remove side cover by pulling on it. It should snap off.
5. Gently remove back cover by pulling on it. It too, should snap off.

Be Careful not to touch anything other than the fan and filter assembly. Many pieces on the back of the scanner will be hot to the touch, and are to be avoided.

VORSICHT: Berühren Sie keine Bauteile, mit Ausnahme des Ventilators und der Filter. Zahlreiche Bauteile an der Rückseite des Scanners sind heiß und dürfen daher nicht berührt werden.

ATTENTION : ne touchez rien d'autre que le module ventilateur et filtre. De nombreux éléments situés à l'arrière du scanner sont chauds, évitez de les toucher.

6. Locate the Fan filter (123816). It is located on the back of the scanner, just under the plastic fan cover.



7. Pull straight back on the plastic fan cover until it comes free from the scanner housing.



8. Remove the filter from inside the cover, and rinse with tap water; replace if necessary.



9. Re-attach the fan cover by applying moderate pressure to each of the four sides, until it snaps back in to place.

Cleaning the Film Track

It is recommended to clean the film track every two weeks, along with the illumination cleaning. Keeping the film track clean will ensure that DX code reading remains reliable. It will also maintain the proper motor speed for each resolution.

1. Remove the film path cover.
2. Position the tip of the compressed air hose on both sides of the film track entrance, and blow.



3. Position the tip of the compressed air hose on both sides of the film track exit, and blow.



4. After cleaning the film track, run the 'Film Track Test' in the client interface program.
5. Verify the scanner is properly reading DX codes by scanning a roll of film.

Surface Cleaning

Clean surfaces with lint-free cloths or mild, non-abrasive spray type cleaners, where the cleaner is sprayed onto the cleaning cloth and not directly onto the equipment. Perform this type of cleaning only with the equipment disconnected and powered down.

Regulatory Information



Warning:

This is a class A product. In a domestic environment this product may cause radio interference in which case the user may be required to take adequate measures.

Note:

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Note:

Japan:

この装置は、情報処理装置等電波障害自主規制協議会（VCCI）の基準に基づくクラスA情報技術装置です。この装置を家庭環境で使用すると電波障害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。

English Translation:

This is a Class A product. In a domestic environment this product may cause radio interference in which case the user may be required to take adequate measures. VCCI-A

Taiwan:

警告使用者:

這是甲類的資訊產品，在居住的環境中使用時，可能會造成射頻干擾，在這種情況下，使用者會被要求採取某些適當的對策。

Site Conditions

Temperature	59-85 degrees Fahrenheit
Humidity	30-75% (non-condensing)
Vibration	Not to Exceed 0.05G RMS 5-200Hz
Ventilation	A 4" clearance above and behind the scanners venti- lation duct is necessary
Lighting	<1000 LUX ambient light
Noise	< 70db(A)

Shipping and Storage Conditions

Temperature: -13 to 140 degrees Fahrenheit

Humidity: 30%-90% (non-condensing)

Fuse Ratings

Only Pakon service technicians are qualified to replace fuses in the F-235 Scanner.

Die Sicherungen im Scanner F-235 dürfen nur von Pakon-Servicetechnikern ausgetauscht werden.

Seuls les techniciens du service d'entretien de Pakon sont qualifiés pour remplacer les fusibles du scanner F-235.

Factory Fuse Rating
3 Amps, 250V

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